How soybean defect classifiers generalize across equipment: a study with open data

Thalles Paul Leandro Mota

Technical report – November 2026

Abstract. *Machines that grade soybeans from images only standardize grading if the model keeps its performance at sites where it was not trained. This report measures that loss with two public datasets (10,343 images), treating three regions as three equipment units. Splitting the photos one by one, instead of by tray, inflated balanced accuracy by 4.4 points. At a unit it had never seen, the model lost 14.2 points, recovered when part of that unit's samples joined the training set. Between datasets of different origin, performance fell to chance; 50 labeled images from the new dataset brought accuracy back to 86.9. Sending the 10% most doubtful beans to human review raised accuracy from 91.4 to 95.2 at a known unit, but helped little at a new one, where the model is wrong with confidence. Of six predictions registered before the tests, 4 were confirmed.*

1. Why I studied this

Automatic grading of soybeans has a bottleneck that a single accuracy figure hides: the same model, moved to another machine, sees different lighting, a different camera angle and beans from another region, and its errors change in ways that are rarely measured. I wanted to look at that problem with the same rigor I use in my research, and to see how far it can be understood and reduced using only open data.

2. Introduction

In Brazil, soybeans are graded under the official standard of the Ministry of Agriculture [MAPA 2007]. In a sample from each truckload, a grader visually separates the damaged beans (moldy, heat-damaged, immature, stink-bug damaged, among others) and weighs each group; when in doubt, the bean is cut open. For Group II soybeans, a load with more than 8% damaged beans falls out of grade. The 2025/26 harvest reached 180.4 million tonnes [Conab 2026], and this separation is repeated for every load received.

Machines that photograph each bean and classify it with a machine-learning model promise speed and the same criterion at every site. That benefit depends on one condition: the model must keep its performance at a site where it was not trained, with another camera, other lighting, another region and another reference grader. This study answers three questions: (i) how much the model loses at a new unit; (ii) why it loses it; and (iii) how much new data, and of what kind, recovers the performance.

3. Data

The SOYPR dataset [Pereira et al. 2024] contains 4,830 images of beans from three regions of the state of Paraná, in seven classes (intact, cercospora, greenish, mechanical damage, stink bug, dirty and moisture). Each bean was photographed three times on a

tray, and each region was labeled by a specialist. In this study, each region plays the role of an equipment unit.

The Mendeley dataset [Lin et al. 2023] contains 5,513 images from another source, in five classes, under a CC BY 4.0 license. The audit found 371 pairs of repeated images (290 byte-identical and 81 recompressed) and one label conflict; I kept one image per group of copies and removed the conflict, leaving 5,141 images. Because the two datasets do not share the same classes, comparisons between them use the common task: intact or defective.

4. Evaluation protocol

Splits. Training and test sets were defined before any model, by physical tray in SOYPR and by group of copies in Mendeley, and frozen with a SHA-256 checksum. Automated tests fail if a tray or a copy appears on both sides. Within each unit, the trays were split at random into two halves 20 times, each split used in both directions.

Reference model. The best model from the SOYPR paper: the 1,000 outputs of a pre-trained ViT-Base network [Dosovitskiy et al. 2021], standardized and classified by an SVM with an RBF kernel. Its hyperparameters (C and γ, 110 pairs) were chosen using training data only; the test set never took part in any choice.

Metrics. Balanced accuracy (the mean of per-class accuracy) as the main metric; per-class accuracy and confusion matrices; weighted F1 only for comparison with the literature. Uncertainty comes from resampling whole trays, not isolated beans.

Predictions. Six predictions, each with a criterion for success, were registered before the tests.

5. Results

5.1. Splitting by photo inflates the result

The three photos of a bean are almost identical. If one lands in training and another in the test set, the model recognizes the bean instead of classifying the defect. Figure 1 compares the two ways of splitting, with the same model and the same tuning.

Splitting photo by photo inflated balanced accuracy by 4.4 points, in all three units; the worst photo-by-photo repetition was above the best one by tray. With honest evaluation, the paper's model reached 71.8 weighted F1 per bean, against 85.1 published; even choosing the hyperparameters on the test set, the ceiling was 74.8. The paper does not report the version of the weights or how the images were prepared, and the honest number was adopted as the reference.

5.2. A unit never seen

Figure 2 compares, on the same test trays, the model trained on the unit itself, the one trained only on the other two, and the general model that includes half of the unit's trays.

The average loss at a unit never seen was 14.2 points (79.4 to 65.2), in the same direction in all 20 repetitions. With part of the unit's trays in training, the general model reached 80.2, on par with the local model. The extreme case was the "dirty" class at unit 2:

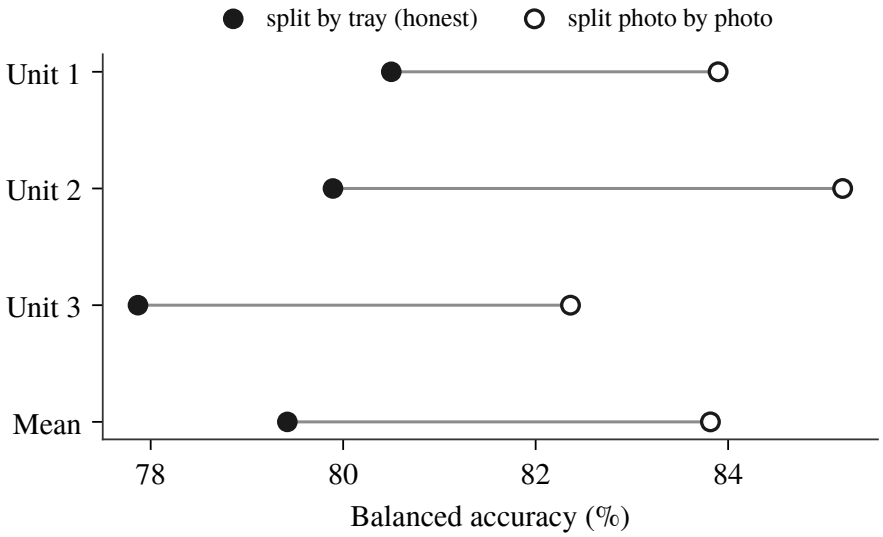


Figure 1. Balanced accuracy with the split by tray and photo by photo (mean of 20 repetitions).

100% accuracy with the local model and 9% without data from that unit. The dataset's authors describe this class as subject to the graders' subjectivity [Pereira et al. 2024], which points to a labeling criterion specific to each region.

5.3. Another image dataset

On the intact-or-defective task, a model trained on one dataset and applied to the other fell to chance (50.1 and 52.3 balanced accuracy, against 93.6 and 94.1 within each dataset), labeling almost every bean as defective. The ability to rank the beans also dropped (AUC of 0.82 and 0.72, against 0.98 to 0.99 within each dataset), so adjusting the decision threshold alone is not enough. Figure 3 shows how much labeled data from the new dataset recovers the performance.

Training only on the new images was the best strategy: 86.9 with 50 images and 90.3 with 200, close to the 94.7 of the whole dataset. Adding the new images to the old dataset was worse, and adjusting the threshold alone stopped at 74.3. On Mendeley, which is harder, the values were 70.1 and 81.1.

5.4. Changing the feature extractor

The SOYPR paper compared five networks, all with an SVM, and the SVM beat a simple neural network. So the classifier was kept and only the network that describes the image was changed (Table 1).

DINOv2 lost less at the new unit (+4.2 points over ViT-Base, ahead in all 20 repetitions), without removing the loss. None of the three networks solved the change of dataset.

5.5. The model's doubt and human review

A committee of the two best networks gives each bean a class and a degree of doubt. Figure 4 shows the effect of sending the most doubtful beans to human review.

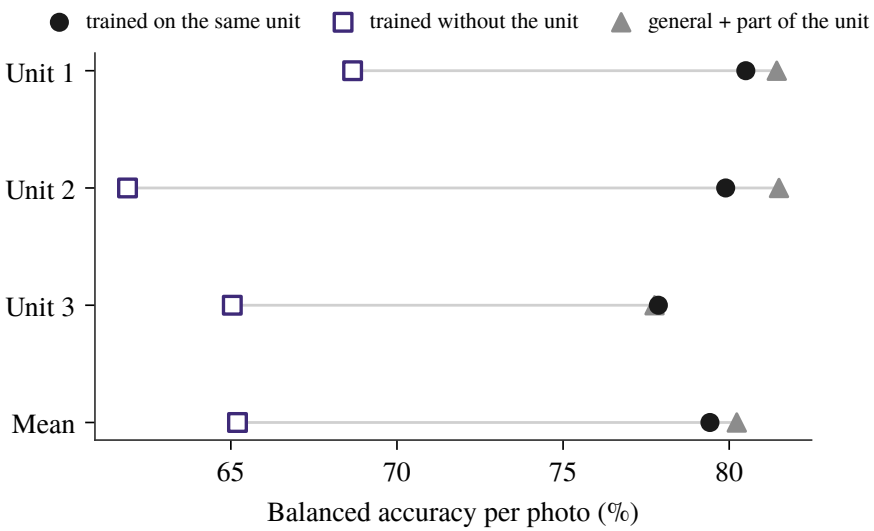


Figure 2. Balanced accuracy per photo on the same test trays of each unit.

Table 1. Balanced accuracy per photo with three pre-trained networks, always with an SVM.

Network	Same unit	Unit never seen	Loss
ViT-Base [Dosovitskiy et al. 2021]	79.4	65.2	−14.2
DINOv2 [Oquab et al. 2024]	80.0	69.4	−10.6
CLIP [Radford et al. 2021]	73.3	60.0	−13.3

At a known unit, reviewing 10% of the beans raised accuracy from 91.4 to 95.2 and caught 56% of the errors. At a new unit, the same review caught only 27% of the errors: the model is wrong with confidence, and its doubt points to errors less well away from home (AUC 0.77) than at home (0.88). An analysis of how hard each bean is [Smith et al. 2014] separated two kinds of error: beans that are ambiguous even at their own unit, and beans that are easy at home but missed elsewhere, typical of the “dirty” class at unit 2.

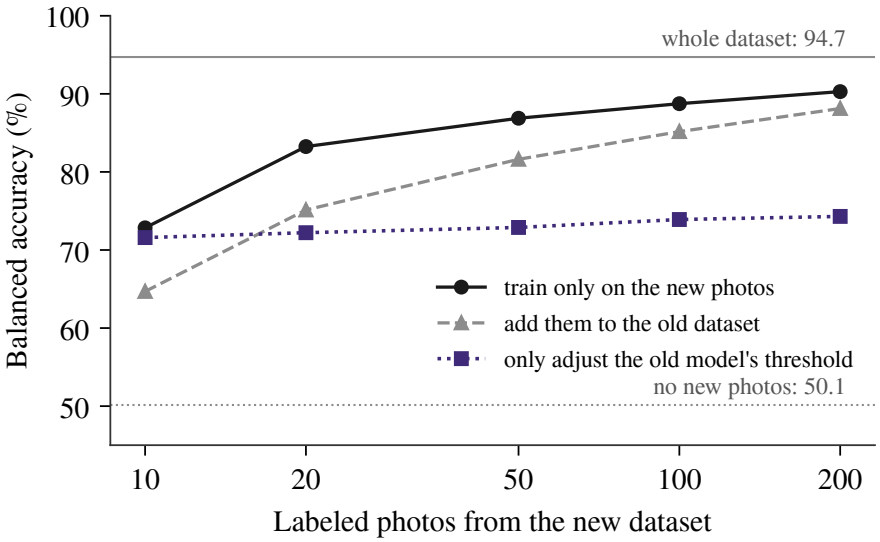


Figure 3. SOYPR as the new dataset: balanced accuracy always measured on trays outside the draw (20 draws per point).

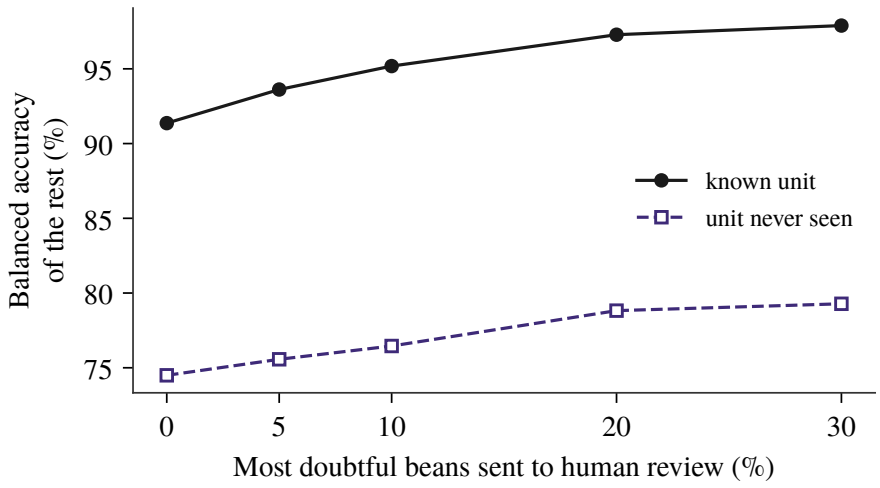


Figure 4. Balanced accuracy per bean for the beans that stay with the model.

6. Predictions registered before the tests

Table 2. Predictions, result and verdict.

	Prediction	Result	Verdict
P1	Splitting by photo instead of by tray inflates the result.	+4.4 points, in all 3 units	confirmed
P2	Without unit 2 in training, "dirty" is the most affected defect.	"dirty": from 100% to 9% accuracy	confirmed
P3	Mechanical damage and moisture damage will have the worst recall in every split.	the two worst in 3 of 9 cases	not confirmed
P4	The drop when changing dataset will be much larger than when changing unit.	43.5 against 0.8 points	confirmed
P5	On Mendeley, a split that ignores the copies inflates the recall of intact beans.	92.9% against 90.8% (4 of 5 parts; small effect)	confirmed
P6	Doubt will be higher at the unseen unit and will concentrate on mechanical damage, moisture and the "dirty" class of unit 2.	higher away: yes; "dirty" at unit 2: yes; mechanical among the most doubtful: no	not confirmed

The two predictions that were not confirmed failed on the same point: mechanical damage was not the hardest class; greenish, moisture and stink bug were.

7. Where it can be used, and recommendations

Table 3. Recommendations by situation, with the values measured in this study.

Situation	Recommendation	Balanced acc.
Known equipment	committee of two networks; review the 10% most doubtful	91.4 → 95.2
New equipment, no data from it	validate before use	76.8
New equipment, with part of its samples	general model with the unit's data	87.8
Another image dataset, none of its images	do not use	chance
Another dataset, 50 or 200 labeled images	train only on the new images	86.9 or 90.3

The values in the first three rows are per bean; those in the last two, per image, on the intact-or-defective task. In order of priority, the study suggests: (i) always evaluate by sample and by unit, before changing the model; (ii) collect and label samples from each new unit before operating it; (iii) use the model's doubt to guide human review at known units; (iv) align the labeling criterion across units; and (v) try newer feature extractors, without expecting them to replace data from the unit.

8. Limitations

The units in this study are photographed regions, not distinct machines; the effects of the equipment, the region and the reference grader cannot be separated. The networks

were used without retraining. The comparison between datasets uses only the intact-or-defective task, because the datasets do not share the same classes. SOYPR images are not reproduced because the dataset does not state a license for reuse.

References

Conab (2026). Acompanhamento da safra brasileira de grãos, v. 13, safra 2025/26, 12º levantamento.

Dosovitskiy, A. et al. (2021). An image is worth 16x16 words: transformers for image recognition at scale. In *ICLR*.

Lin, W. et al. (2023). Soybean image dataset for classification. *Data in Brief*, 48:109300.

MAPA (2007). Instrução Normativa n. 11, de 15 de maio de 2007: Regulamento Técnico da Soja.

Oquab, M. et al. (2024). DINOv2: learning robust visual features without supervision. *Transactions on Machine Learning Research*.

Pereira, G. M. L., Foleis, J. H., Brito Jr., A. S. and Bertolini, D. (2024). A database for soybean seed classification. In *37th SIBGRAPI*. IEEE.

Radford, A. et al. (2021). Learning transferable visual models from natural language supervision. In *ICML*.

Smith, M. R., Martinez, T. and Giraud-Carrier, C. (2014). An instance level analysis of data complexity. *Machine Learning*, 95:225–256.